Reading Notes For Week 9

- **Distinction Between Associational and Causal Inferences:**
    - Associational inference deals with relationships between variables as observed in data, without implying causation.
        - Computing covariance is association analysis, not causal analysis.
    - Causal inference aims to establish a cause-and-effect relationship, requiring a more structured analytical framework.
- **The Rubin Causal Model:**
    - Introduces a formal model for causal inference that differentiates between potential outcomes under different treatments or interventions.
        - Learned in DS-GA-1002, potential outcomes contain the outcomes for all individuals under certain treatment conditions.
        - Want to investigate whether a drug is effective, PO1 denote the potential outcome of an individual that take the drugs, PO0 denote the potential outcome of an individual that don't take the drugs.
        - Compute the E(PO1) - E(PO0) and see whether it is positive.
            - If it is, then we say taking the drug causes the potential outcome(e.g. longevity) to increase.
    - Each unit (e.g., individual in a study) has potential outcomes that could be observed under each possible treatment, but only one outcome is observable since each unit can receive only one treatment.
        - The outcome not observed is called counterfactual.
- **Fundamental Problem of Causal Inference:**
    - It's impossible to observe all potential outcomes for a unit simultaneously—only the outcome under the actual treatment is observed, not the counterfactuals.
    - This problem highlights the inherent challenge in directly measuring causal effects.
- **Strategies for Overcoming the Fundamental Problem:**
    - Emphasizes the importance of randomized experiments, where treatment assignment is random, to estimate causal effects.
    - Discusses statistical adjustments and assumptions (e.g., independence, homogeneity) that allow for causal inference from observational data.
- **Average Causal Effect (ACE):**
    - Defined as the average difference in outcomes between units under one treatment versus another, across the population.
    - I think it is central to understanding the overall impact of a treatment or intervention.
    - Can be related to Average Treatment Effect(ATE).
- **Assumptions for Causal Inference:**
    - **Independence:** The treatment assignment is independent of potential outcomes. $p(PO)=p(PO|t)$

        - This assumption only holds when the treatment is randomized or when we spot the confounding factor and adjust for that.
    - **Conditional Independence (From DS-GA-1002):** Conditioned on the confounding factor, there is no systematic difference between treatment and control group. p(PO|t,a) = p(PO|a)
    - **Homogeneity/Constant Effect:** The treatment effect is constant across units, simplifying the inference of causal effects.
- **Critique of Traditional and Philosophical Views on Causation:**
    - Evaluates how traditional statistical and philosophical approaches to causation align or conflict with the Rubin Causal Model.
    - Discusses the limitations of associational statistics in inferring causation and the need for a causal inference framework.